BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: 25-26J-089

Project Title: Data driven analysis of university past paper & lecture content for curriculum optimization

1. Introduction
   1.1. Background:

In university education, assessments play a critical role in evaluating student understanding and guiding learning outcomes. However, students often face difficulties in preparing effectively for examinations due to the lack of clear alignment between lecture materials, curriculum objectives, and past examination patterns. Lecture notes are usually extensive and unstructured, while past examination papers span multiple years, making it challenging to manually identify important topics, recurring patterns, and assessment focus areas.

Traditional methods of analyzing academic materials rely heavily on manual effort by students and educators. Such approaches are time consuming, prone to inconsistency, and limited in their ability to extract meaningful insights from large volumes of academic data. As a result, students may focus on less relevant content while overlooking topics that are frequently assessed, leading to inefficient preparation and increased academic stress.

Advancements in data science and natural language processing provide new opportunities to address these challenges. By applying data driven analytical techniques to past examination papers and lecture materials, it is possible to systematically identify topic relevance, assessment trends, and structural characteristics of exams. This research explores the use of academic data as a foundation for developing intelligent educational support mechanisms that enhance exam preparation, improve learning efficiency, and support evidence based decision making in higher education.

   1.2. Research Problem:

- How to align unstructured DMS lecture materials with syllabus modules and examination requirements to support efficient exam preparation?
- How to prioritize and summarize lecture content based on historical exam trends to generate exam-focused study resources?
- How to provide multimodal learning support, including text, image, and voice-based explanations, to improve conceptual understanding?
- How to enable curriculum-aligned, Bloom's Taxonomy–based question answering and Continuous Assessment (CA) guidance?
- How to continuously improve the accuracy and relevance of learning support using student feedback and reinforcement mechanisms?

1.3. Objectives:

It Number	Objective	Objective number
IT22891440	1. Analyze past examination papers to identify recurring question patterns, topic distributions, mark allocations, and structural formats commonly used in university examinations. 2. Generate authentic model examination papers that closely reflect historical exam structures, topic coverage, and difficulty balance, ensuring alignment with curriculum expectations. 3. Extract and organize lecture content to produce concise, topic focused short notes that directly support the questions included in the generated model papers. 4. Ensure alignment between model exam questions and corresponding short notes so that students can easily revise relevant concepts based on exam focused requirements.	1

	5. Improve the effectiveness of exam preparation by reducing manual effort and providing structured, data driven learning materials that support focused revision.	
IT22171306	1. **Lecture Note Summarization** To generate concise and topic-specific summaries from DMS lecture notes based on user-selected titles, enhanced with image-based explanations and voice narration for improved conceptual understanding. 2. **CA (Continuous Assessment) Guidance System** To assist students by providing relevant documentation links, video resources, and automatically creating assignment deadlines in a calendar to support effective academic planning. 3. **Bloom's Taxonomy - Based Q&A Module** To enable question answering from lecture notes across different cognitive levels of Bloom's Taxonomy	2

	(remember, understand, apply, analyze, evaluate, create), supporting structured learning and self-assessment. 4. **Reinforcement Learning via Student Feedback** To improve system performance by allowing students to comment and provide feedback, which is used as a reinforcement signal to iteratively enhance summary quality, answer relevance, and guidance accuracy. 5. **System Evaluation and Reliability** To evaluate the effectiveness of the system using relevance checks, feedback-based reward signals, and standard NLP evaluation metrics to ensure reliability and continuous improvement.	
IT22598196	1. **Comprehensive Data Collection and Preparation** To collect past DMS examination papers and lecture notes, and perform data cleaning, normalization, and	3

	preprocessing to build a structured, high-quality dataset suitable for domain-specific analysis. 2. **Automated Question Difficulty Prediction** To analyze DMS examination questions using Bloom's Taxonomy in order to accurately predict and assign multiple levels of question difficulty. 3. **Question Categorization and Repository Development** To classify and organize DMS questions into meaningful categories based on topics, cognitive levels, and difficulty, and to create a searchable and reusable question repository. 4. **Personalized and Adaptive Feedback Generation** To provide AI-driven, personalized feedback and topic recommendations to students by analyzing their essay responses and performance trends, enabling continuous improvement through adaptive learning mechanisms. 5. **User-Friendly Interface and System Support** To design and implement an intuitive, accessible,	

	and learner-friendly user interface that supports diverse learners and ensures effective interaction with the AI-based learning support system.	
IT22113672	**1.Data Collection & Preprocessing:** Collect, preprocess, and clean a decade of SLIIT DMS exam papers (MCQs) and lecture materials using OCR, tokenization, and error correction techniques to create a high-quality dataset suitable for NLP analysis **2.Syllabus Module Classification:** Manually or semi-automatically label a portion of the questions and lecture content with their relevant syllabus module tags. Use Sentence-Transformer to generate semantic embeddings that capture contextual meaning. **3.Topic Importance Analysis:** Using classified questions, analyze the frequency and weight of each topic.This identifies high-frequency topics, enabling the system to prioritize high-impact areas based on historical exam trends and question density. **4.Personalized Time-Aware Study Plan:** Collect student inputs (exam date and daily study hours). The scheduling algorithm allocates study time to each topic based on importance	4

	and remaining days, producing a personalized, efficient topic-wise study schedule. **5.Recommend the MCQ Questions :** Implement GraphRAG-based embedding similarity search to recommend the most relevant past MCQ exam questions per topic, tagged with difficulty level and syllabus module. Integrate into a user-friendly interface	

2. Data Exploration
   2.1. Data Collection

Data for this research were collected from multiple academic and educational sources to support curriculum optimization and personalized learning. The dataset was compiled using SLIIT's internal academic platforms and external educational websites to ensure comprehensive coverage and reliability.

The primary sources included the **SLIIT eLibrary, from which past examination papers spanning a period of ten years were collected**, and the **SLIIT Courseweb platform, from which eight lecture slides and associated course materials were obtained**. In addition, external educational websites were utilized to collect supplementary multiple-choice questions (MCQs) to enrich the dataset.

The integration of these data sources enables detailed analysis of syllabus coverage, topic-wise question distribution, and examination trends, thereby facilitating the development of a data-driven personalized learning support system.

2.2. Dataset Description:

(Describe datasets, including sources, size, and key attributes)

Data source	Description	Resource	Key attributes
1. Past Papers	Historical university exam papers for the DMS module, used to analyze structural patterns, topic frequencies, question types, and assessment trends. These enable the generation of model exam papers, difficulty predictions, and personalized study recommendations by identifying high-frequency topics and redundancies.	Past papers	1.Includes questions(e.g., MCQs, essays, structured problems) 2.Mark distributions and section formats 3.Topic coverage(e.g., database design, SQL queries, normalization) 4.Question difficulty levels and historical trends 5.Metadata such as year, exam type, and module alignment
2.Lecture Slides	Official lecture slides and notes for the DMS module, serving as reference material for semantic mapping to exam topics, content summarization, flashcard generation, and personalized short notes. These help bridge gaps between taught content and assessments.	Lecture Slides	1. Explanations of concepts (e.g., slides on data models, relational algebra) 2. Visual elements such as diagrams, tables, and code snippets 3. Topic hierarchies and key terms for NLP processing 4. Content depth aligned with syllabus modules 5. Formatting considerations for

			extraction (e.g., PDF/DOCX, slide-wise organization)
3. External Educational Website	Supplementary multiple-choice questions (MCQs) collected from reputable educational websites to enrich the dataset and improve coverage of diverse question formulations	MCQ's Website	Large collection of topic-wise multiple-choice questions related to Database Management Systems Consistent and standardized question–option formats suitable

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1	×		×	×
Data source 2	×	×	×	×
Data source 3				×

3. Methodology

3.1. Data Preprocessing:

(Mention data transformation techniques done in each dataset for each objective.)

Ex:

Data Cleaning, Data Normalization, Data Standardization, Data Encoding (e.g., One-Hot Encoding, Label Encoding), Handling Missing Data (e.g., Imputation or Removal), Data Aggregation, Feature Engineering, Outlier Detection and Handling, Data Scaling, Data Discretization, Dimensionality Reduction (e.g., PCA), Date/Time Transformation, Data Integration (Merging or Joining), Data Mapping, Data Type Conversion.

Objective 1

	Transformation technique							
	Data Cleaning	Data Normalization	Data Encoding	Feature Engineering	Data Aggregation	Data Mapping	Data Type Conversion	Manual Correction
Data source 1	×	×	×	×	×	×	×	Manually verified extracted questions, corrected OCR related text errors, and validated mark distributions to ensure accuracy
Data source 2	×	×	×	×	×	×	×	Performed a light manual verification to confirm that the automatically extracted keywords and topic labels were semantically relevant to the lecture content and syllabus,

								without altering the underlying dataset.

Objective 2

	Transformation technique								
	Data Cleaning	Data Normalization	Data Encoding	Feature Engineering	Data Aggregation	Data Mapping	Data Type Conversion	Date/Time Transformation	Manual Correction
Data source 2	×	×	×	×	×	×	×	×	Manually reviewed a small sample of generated summaries and multimodal explanations to verify conceptual correctness and syllabus alignment, without making manual changes to the dataset.

Objective 3

	Transformation technique							
	Data Cleaning	Data Normalization	Data Encoding	Feature Engineering	Data Aggregation	Data Mapping	Data Type Conversion	Manual Correction
Data source 1	×	×	×	×	×	×	×	Review & correct misclassified Bloom's levels (especially borderline cases) - Remove false-positive questions
Data source 2	×	×	×	×	×	×	×	Manually verify/correct important topic names that were missed or wrongly capitalized by extraction

Objective 4

	Transformation technique							
	Data Cleaning	Data Normalization	Data Encoding	Feature Engineering	Data Aggregation	Data Mapping	Data Type Conversion	Manual Correction
Data source 1	×	×	×	×	×	×	×	Review and filter false-positive questions using pattern validation and content checks, remove invalid entries
Data source 2	×	×	×	×	×	×	×	Manually verify important topic names that were missed or wrongly capitalized by extraction

Data source 3	×	×	×	×	×	×	×	Manually reviewed ambiguous or low-quality questions, removed misleading or poorly worded MCQs.

3.2. Scalability

The existing datasets, consisting of 15 past DMS examination papers and 8 sets of lecture slides, are sufficient in size to support the development, training, validation, and demonstration of the current system components including question classification, difficulty-based selection, answer evaluation, personalized feedback, and adaptive learning behavior.

This volume of data enables reliable pattern identification (topic frequency, cognitive demand trends, and content coverage gaps) and provides a solid foundation for prototype-level performance within the scope of a final-year research project.

Importantly, the overall system design remains highly scalable and modular. Should a new module be added in the future the architecture can easily accommodate expansion by:

- Ingesting more years of past papers
- Incorporating lecture materials from additional subjects
- Processing larger volumes of anonymized student responses

without requiring major structural redesign. The preprocessing pipeline, question pooling, evaluation logic, and reinforcement learning adaptation mechanism are inherently data-agnostic and can scale with increased data volume through simple extension of the data directory and batch processing.

Therefore, while the current dataset sizes are adequate and appropriate for the defined project objectives, they also provide a strong, extensible foundation suitable for future module additions and potential institutional-level deployment.

3.3. Feature extraction

Objective 1

1. Key Features Evaluated

- Question text content extracted from past examination papers, including main questions and sub questions.
- Mark allocation values associated with each question and sub question.
- Topic specific keywords extracted from both past exam questions and lecture notes using statistical and semantic relevance measures.
- Structural features such as question position, number of sub questions, and section wise distribution.
- Lecture concept keywords representing definitions, explanations, and procedural steps relevant to exam topics.

2. Feature Transformation

- Textual content from past papers and lecture notes was tokenized and converted into numerical representations using vector based text encoding techniques.
- Extracted keywords were transformed into feature vectors to support similarity based comparison between exam questions and lecture topics.
- Mark values and question indices were converted into structured numerical features for pattern analysis.
- Aggregated feature representations were created to capture recurring exam patterns across multiple past papers.

Objective 2

1. Key Features Evaluated

    - Lecture text content extracted from DMS lecture slides after conversion to PDF, including topic headings, subtopics, definitions, and explanatory points.
    - Topic-wise textual segments derived from lecture titles to support title-based summarization and retrieval.
    - Diagram images extracted from lecture slides representing conceptual models, architectures, and workflows relevant to the subject matter.
    - Semantic metadata such as source document, page number, content type (text, table, image), and image identifiers.
    - User query features derived from student-selected titles and questions to enable context-aware retrieval and generation.

2. Feature Transformation

- Textual lecture content was tokenized and transformed into dense semantic embeddings using transformer-based text encoding models for retrieval and similarity matching.
- Extracted diagram images were converted into visual embeddings using a CLIP-based image encoder to support multimodal explanation.
- Text and image embeddings were projected into a common fixed-dimensional vector space and normalized to enable unified similarity comparison.
- Metadata features were structured and attached to each content chunk to facilitate accurate mapping between text, images, and generated responses.
- Aggregated feature representations were created by combining multiple retrieved lecture chunks to form a coherent context for summarization, question answering, and CA guidance.

Objective 3

1. Key Features Evaluated

- Detected specific action verbs (e.g., "define", "explain", "evaluate") using keyword pattern matching directly determines preliminary difficulty bucket (easy/medium/hard).
- Measured word count, sentence count, and average sentence length longer and more compound questions typically indicate higher cognitive demand.
- Counted frequency of DMS core terms (e.g., "normalization", "ER diagram", "SQL injection", "transaction") higher density correlates with topic importance and potential difficulty.
- initial semantic overlap between question text and extracted lecture topics (via keyword co-occurrence) questions with low traceability tend to require deeper inference and are evaluated as harder.
- Presence of interrogative patterns, multi-part structures, and requirement for explanation/justification helps differentiate descriptive (easier) from analytical/evaluative (harder) questions.

2. Feature Transformation

- Mapped Bloom's verb-based classification to discrete labels: "easy", "medium", "hard" used directly for stratified question selection and RL state representation.
- Scaled question length and keyword density values to a 0–1 range (min-max normalization) prevents feature dominance and improves consistency across different exam years.
- Converted presence of specific Bloom's verbs and DMS keywords into binary or multi-hot vectors facilitates fast filtering and prompt context enrichment.

- Grouped and bucketed all processed questions into three separate lists (easy/medium/hard) based on combined feature scores enables efficient random stratified sampling during adaptive questioning.
- Aggregated and cleaned lecture-extracted topics into a unique, standardized reference list transformed into a compact context string for consistent evaluation and study recommendation generation.

Objective 4

1.Key Features Evaluated

- Semantic similarity between exam questions and lecture content is measured using Sentence- Transformer (all-MiniLM-L6-v2) embeddings to accurately determine syllabus module alignment.
- Topic recurrence and importance are evaluated by analyzing question frequency and TF-IDF scores across ten years of DMS exam papers.
- The relevance of questions to high-priority syllabus topics is assessed to identify areas with greater exam impact.
- Student-specific constraints such as exam date and available daily study hours are evaluated to support time-aware personalization.
- Semantic closeness between topics and past exam questions is analyzed using GraphRAG based similarity to enable targeted practice question recommendation.

2. Feature Transformation

- Sentence- Transformer embeddings are generated for exam questions, lecture slides, and syllabus topics to create a unified semantic feature space.
- Topic frequency and TF-IDF values are normalized to comparable scales to ensure balanced contribution in priority scoring.
- Semantic similarity scores are transformed into ranked topic importance values used directly in study plan generation.
- Graph-based representations are constructed by linking questions, topics, and syllabus modules to support efficient GraphRAG retrieval.
- Final transformed features are integrated into structured inputs for personalized study scheduling and practice question recommendations.

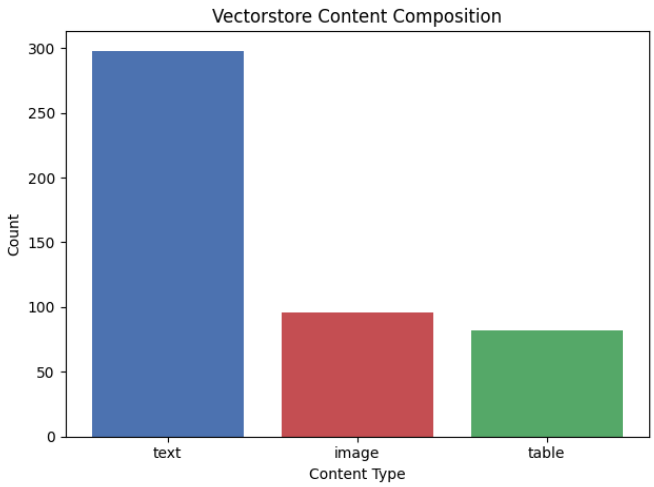
4. Modelling and Results
   4.1. Key Insights:

Objective 1

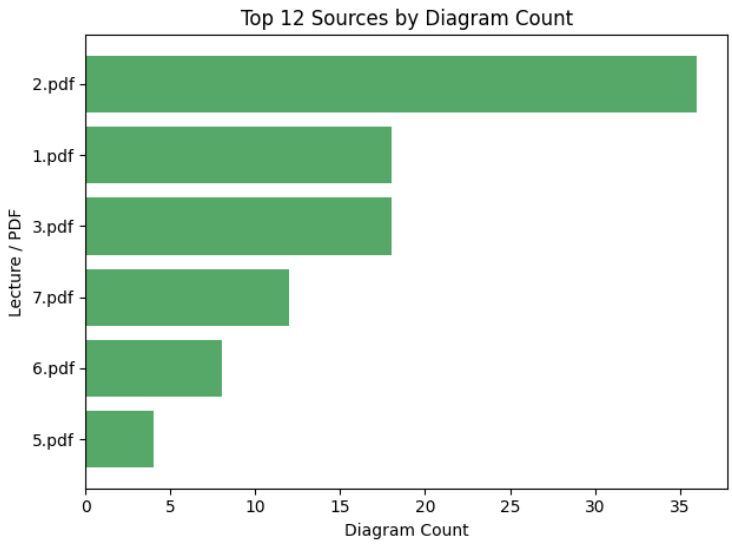
- Analysis of past examination papers showed consistent exam structures across multiple years, with similar question positions carrying comparable mark allocations.
- Topic frequency analysis revealed that a small set of core Database Management Systems topics appeared repeatedly in exams, indicating their high importance.
- Mark distribution patterns indicated that higher mark questions were commonly associated with application based and problem-solving topics.
- Examination questions were observed to assess multiple cognitive levels, including recall, understanding, and application, rather than focusing only on memorization.
- Comparison between lecture notes and past papers showed strong overlap for foundational topics, while some lecture content appeared less frequently in exams.
- Lecture sections with higher conceptual density demonstrated stronger alignment with structured exam questions.
- Overall findings confirmed that past exam papers follow stable patterns and that lecture materials can be selectively aligned to support exam focused short notes and model paper generation.

Objective 2

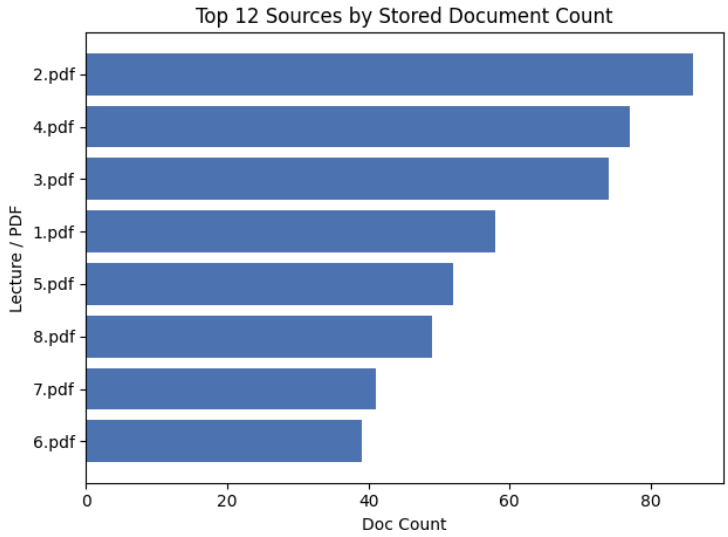
- Lecture slides primarily contained concise bullet-point text, emphasizing definitions, key concepts, and procedural steps rather than full descriptive paragraphs.

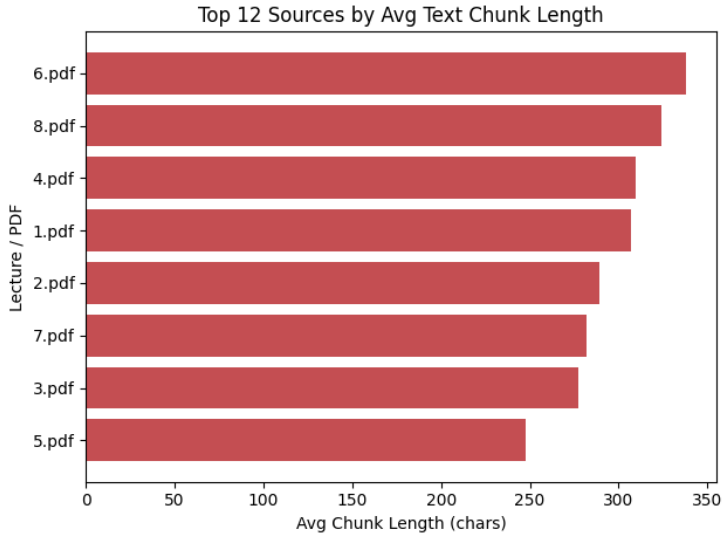


- Topic-wise analysis showed that certain core concepts recur across multiple lectures, indicating their importance for summarization, question answering, and assessment preparation.
- Diagram analysis revealed that visual representations such as architectures, workflows, and conceptual models are frequently used to explain complex topics, highlighting the need for multimodal explanations.



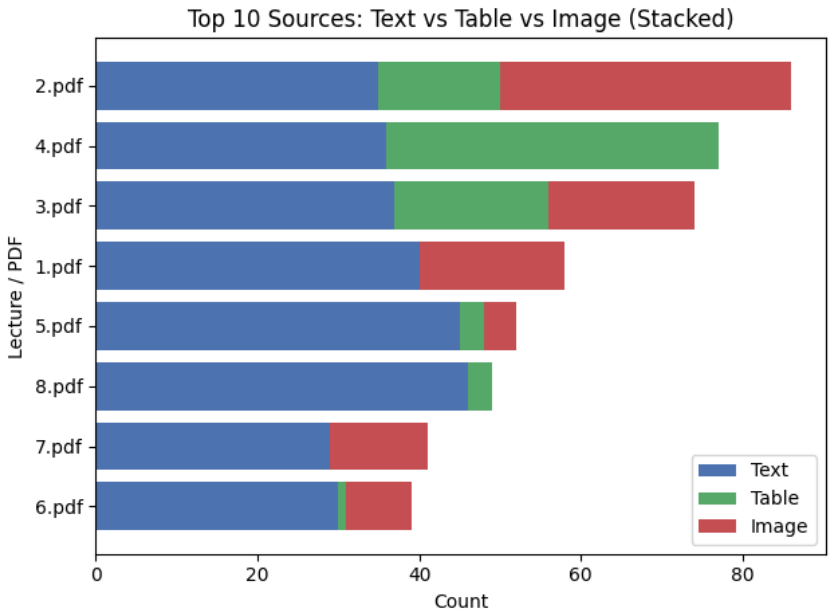
- Textual embedding analysis demonstrated that semantically related lecture topics cluster closely in vector space, validating the effectiveness of transformer-based embeddings for topic-level retrieval.
- Image embedding analysis showed that diagram-related images are more semantically aligned with conceptual queries compared to decorative or illustrative images, supporting the use of diagram filtering.
- Retrieval experiments indicated that smaller text retrieval sets produce concise and focused responses, while extended image retrieval improves visual explanation coverage.
- Aggregated context analysis confirmed that combining multiple relevant lecture chunks improves summary coherence and answer completeness compared to single-chunk retrieval.





- Query pattern analysis revealed that students frequently use short title-based queries, reinforcing the need for title-aligned summarization and retrieval mechanisms.

- Multimodal output evaluation showed that image- and voice-supported explanations improve conceptual clarity, particularly for abstract or process-oriented topics.



- Feedback trends indicated that grounded, syllabus-aligned responses were consistently rated more useful, validating the RAG-based approach.

Objective 3

- The system successfully extracts and identifies important lecture topics and key concepts automatically from provided lecture PDFs, with high accuracy after cleaning and topic aggregation steps, enabling reliable context for answer evaluation and study recommendations.
- Accurate extraction and validation of essay-style questions from past question paper PDFs was achieved using advanced text processing and OCR fallback techniques, resulting in a clean and usable pool of validated questions across multiple years of papers.
- Question classification into Easy, Medium, and Hard levels using Bloom's Taxonomy keywords proved effective.
- Medium-difficulty questions form the largest and most stable portion of assessments over multiple years, making them the ideal default level for starting adaptive question generation and ensuring most learners begin with appropriately challenging practice.
- A noticeable increasing trend in Hard-level questions was observed in more recent examination papers, highlighting the growing importance of higher-order thinking skills and the need for the system to dynamically adapt difficulty progression.
- Core Database Management Systems topics appear repeatedly across the majority of past papers, confirming their critical importance these topics are prioritized for adaptive question generation and targeted topic recommendations.

- There is a clear traceability gap for Hard questions they show noticeably weaker direct overlap with lecture content compared to Easy and Medium questions indicating that higher-order questions often require inference beyond direct lecture material; this justifies the system's mechanism for recommending specific lecture sections when learner performance is low.
- Overall Findings The analysis confirms that the implemented system effectively:Automatically extracts and structures key lecture concepts and validates exam questions with high reliability,Classifies questions meaningfully using Bloom's Taxonomy for adaptive difficulty selection,Dynamically generates appropriate questions based on current learner performance,Provides accurate evaluation, detailed feedback, precise topic recommendations, and updates learning difficulty accordingly

Objective 4

- The system effectively extracts and processes unstructured text from lecture slides and exam paper PDFs using advanced preprocessing techniques, including cleaning, tokenization, and OCR fallbacks, resulting in high-fidelity content ready for semantic analysis and topic mapping.
- Semantic classification via Sentence- Transformer (all-MiniLM-L6-v2) embeddings accurately maps exam questions and lecture materials to syllabus modules (e.g., Normalization, SQL Queries)
- Frequency analysis across a decade of SLIIT DMS exam papers identified recurring high-priority topics, enabling data-driven prioritization and visualizations like bar charts that highlight Pareto patterns for focused study.
- Graph-based similarity searches with GraphRAG and centroid embeddings revealed strong correlations between lecture topics and practice questions.
- Personalized study plan generation dynamically allocates time based on question percentages and user inputs (e.g., exam dates, daily hours), showing trends where top-priority lectures cover exam questions.
- A noticeable gap in direct overlaps for underrepresented lecture sections was observed, indicating that while core topics align well, lesser-frequent areas require enhanced recommendations, justifying the system's modular scalability to other SLIIT modules.
- There is a clear trend of increasing question density in foundational lectures over the years, confirming the need for adaptive scheduling that prioritizes high-impact topics and integrates progress trackers for better engagement and long-term retention.
- Overall Findings: The analysis confirms that the implemented system effectively: Automatically extracts and semantically classifies content using NLP techniques, Identifies recurring patterns and generates personalized, time-aware study plans, Recommends relevant practice questions via graph-based methods, Ensures privacy-compliant scalability,

and Ultimately enhances exam preparation, curriculum alignment, and academic performance for DMS students.

4.2. Challenges Faced During Data Analysis

Objective 1

- Past examination papers were not consistently formatted, making automated text extraction and structural identification more difficult.
- Several papers contained misaligned text, mixed layouts, and embedded diagrams, which affected accurate segmentation of questions and sub questions.
- The dataset included both digitally generated and scanned examination papers, introducing inconsistencies in text quality.
- Additional preprocessing and validation steps were required to ensure that extracted content preserved the original exam structure and academic meaning.

Objective 2

- Lecture slides contained unstructured and fragmented content such as bullet points, incomplete sentences, and mixed formatting, making consistent text extraction and segmentation difficult.
- Extracted images varied in quality, size, and relevance, requiring additional filtering to identify diagrams useful for conceptual explanation.
- Aligning textual content with relevant diagrams was challenging due to the absence of explicit links between text and images in lecture slides.
- Generating text and image embeddings was computationally intensive, requiring limitations on batch size, parallelism, and embedding dimensions.
- Selecting optimal retrieval parameters involved trade-offs between context completeness, noise, and computational cost.
- Student queries differed widely in specificity and structure, making it challenging to ensure consistent retrieval and generation performance.
- Objective evaluation of generated summaries and explanations was difficult due to the subjective nature of educational content, limiting the use of fully automated evaluation metrics.

Objective 3

- Many past examination papers contained misaligned text, inconsistent layouts, varying font sizes, and embedded diagrams or tables, which significantly complicated accurate automatic segmentation of essay questions and sub-questions during extraction.
- The dataset included a mix of digitally generated PDFs and older scanned examination papers, leading to inconsistent text quality,character recognition errors and missing or garbled text in scanned documents.

- Lecture slides often featured heavy use of bullet points, short phrases, repeated slide titles, presenter notes, and non-linear formatting, making it difficult to reliably extract coherent, meaningful topic headings and key concepts without manual intervention.
- The Bloom's Taxonomy keyword classification occasionally misclassified borderline or context-dependent questions,requiring additional manual review and correction of a subset of questions to ensure reliable difficulty level assignment.
- The presence of multi-part essay questions with nested sub-questions and varying mark allocations created structural ambiguity, which affected consistent question isolation and validation, necessitating custom post-processing logic and repeated validation passes.

Objective 4

- Scanned past examination papers exhibited inconsistent OCR quality, resulting in missing words, fragmented sentences, and noise within the extracted text, which initially affected semantic content processing.
- Certain DMS syllabus sections were underrepresented in past examinations, causing imbalance in frequency-based topic importance analysis.
- During semantic similarity measurement, higher-order conceptual questions showed limited direct keyword overlap, requiring deeper contextual inference and making similarity scoring more complex.
- Processing Sentence-Transformer embeddings and GraphRAG-based similarity searches on large datasets introduced memory and computational constraints, necessitating the use of batching and scalable processing techniques.
- Establishing meaningful graph relationships between questions, topics, and syllabus modules required iterative tuning to ensure relevant practice question recommendations.